

A Pitaya Dye-Sensitized Solar Cell Monitor for Environmental Sunlight Intensity Detection

Cheng-Ta Chiang[✉], Senior Member, IEEE, and Liang-Yu Chien

Abstract—For measuring environmental sunlight intensity, a novel pitaya dye-sensitized solar cell (DSSC) monitor was utilized in this paper. A pitaya (botanical name: *Hylocereus undatus*) DSSC sensor and a proposed sensor transducer are consisted together. A linear conversion of environmental sunlight intensity into a frequency output is performed. The novel voltage-to-current conversion technique has digital calibrations to completely eliminate nonlinear terms coming from the analog threshold variations of each metal-oxide-semiconductor field-effect transistor (MOSFET). Measurements showed that the operations of the pitaya DSSC sensor and the proposed sensor transducer were correctly verified. With the proposed technique, the highest sensitivity of -2.7326 MHz/V was guaranteed. The sunlight intensity ranged from 67.775 to 190.45 W/m², with the corresponding frequency ranging from 2.83 to 4.49 MHz. A maximum linear error of 0.79% was obtained. Sensitivity of -10.997 (kHz/W/m²) was measured, and the chip area was 1.49×1.49 mm². Eventually, an experiment was conducted for monitoring orchid environmental sunlight intensity. The results proved that the proposed chip was suitable for measuring sunlight intensity inside greenhouses.

Index Terms—DSSC sensor, monitor, sunlight intensity, orchid.

I. INTRODUCTION

RECENTLY, concerns regarding energy saving and carbon reduction have become more pressing. In the context of such concerns, solar cells have the advantages of green power generation and environmental friendliness. However, traditional solar cells are manufactured from silicon materials. Traditional manufacturing requires expensive equipment, factories, and facilities. Thus, scholars have investigated dye-sensitized solar cells, a new type of solar cell [1]–[25]. DSSC manufacturing uses cheap production materials and low-cost apparatuses. Although various DSSC studies [1]–[25] have been published, none have incorporated any high-sensitivity sensor transducers as a sunlight intensity monitor. Thus, a sunlight intensity monitor realized by a pitaya DSSC sensor and sensor transducer is proposed. Various DSSC sensors could be conveniently matched to the proposed sensor transducer, and the present condition of sunlight intensity is linearly converted to a frequency output. Owing to the digital output [26], it has lower noise sensitivity than conventional

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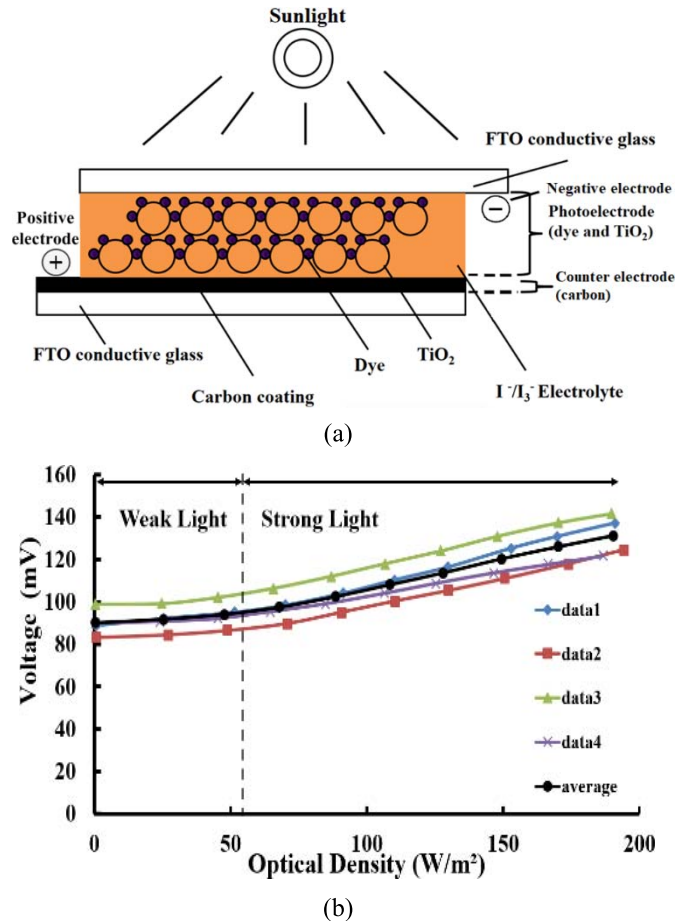


Fig. 1. (a) The manufactured process of a pitaya DSSC. (b) Measured results of the pitaya DSSC sensor.

DSSC sensors with analog outputs. Moreover, the proposed sensor transducer can easily detect sunlight intensity and provide digital output for backend Internet of Things (IoT) devices. To eliminate nonlinear terms coming from the threshold variations of each metal-oxide-semiconductor field-effect transistor, a voltage-to-current conversion with accuracy and digital calibrations was developed together.

The functionalities of the proposed monitor were examined by measurements. The sunlight intensity ranged from 67.775 to 190.45 W/m²; the corresponding frequency ranged from 2.83 to 4.49 MHz. This range was suitable for the light energy transfer to the plant canopy in a Taiwan's orchid greenhouse [27]. A maximum linear error of 0.79% was measured, and the chip area was 1.49×1.49 mm². An experiment for

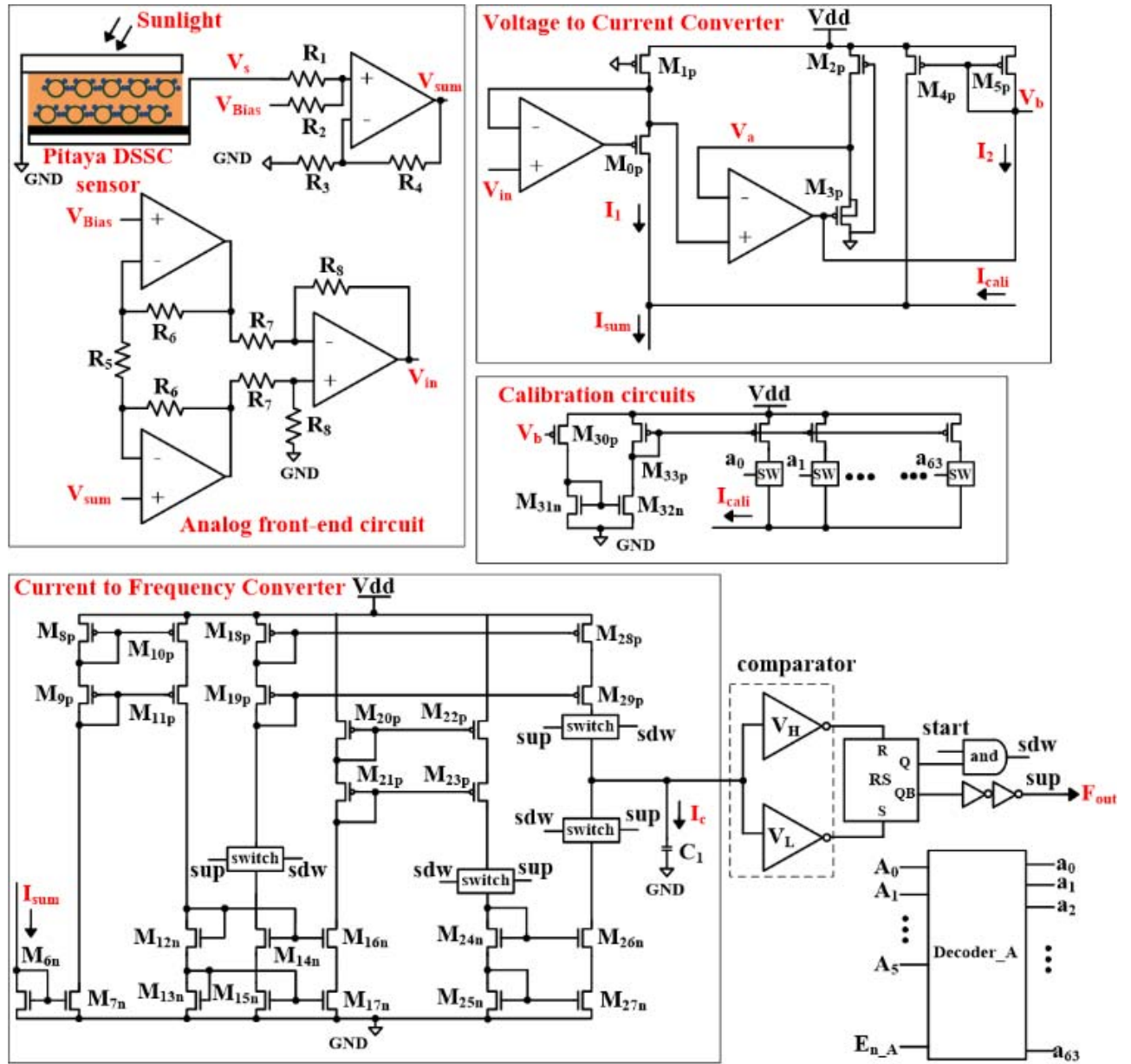


Fig. 2. The circuit methodology.

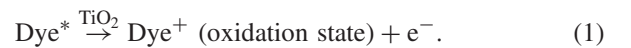
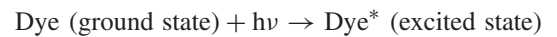
monitoring the intensity of sunlight on orchids was conducted. The results proved that the proposed chip was suitable for measuring sunlight intensity inside a greenhouse.

The design methodology and simulation results of the proposed chip in Section II are showed. Measurements are presented in Section III. A conclusion and future research are discussed in Section IV.

II. DESIGN METHODOLOGY

The manufacturing process of a pitaya DSSC sensor is shown in Fig. 1(a). First, water and titanium dioxide (TiO_2) were mixed to form a colloidal material. The colloidal material was spin-coated on a fluorine-doped tin oxide conductive (FTO) glass substrate. Thereafter, the FTO glass substrate was annealed at 300°C for 30 minutes. The FTO glass substrate was immersed into a mixture of pitaya juice and alcohol for 20 minutes. A separate FTO glass substrate was

also coated with carbon molecules. The carbon molecules were generated by a burning candle. This separate FTO glass was thus used as a counter electrode. Finally, the pitaya DSSC sensor was completed by encapsulating a thin layer of I_3^-/I^- electrolyte between the dye-sensitized TiO_2 and the carbon-coated FTO glass substrate. The mechanism of pitaya DSSC sensor [28] was



When the sunlight was incident on the pitaya DSSC sensor, the photosensitizers (Dye) were excited from the ground state to excited state. The excited electrons were injected into the conduction band of the TiO_2 electrode. This resulted in the oxidation of the photosensitizer. Finally, the electrons occurred on the counter electrode. Next, the output voltage of the

pitaya DSSC sensor was measured. As illustrated in Fig. 1(b), it was linear versus its corresponding optical intensity from 67.775 to 190.45 W/m². Based on this measurement, the pitaya DSSC sensor was employed in the subsequent circuit design.

The circuit methodology is illustrated in Fig. 2. After a voltage generated by the pitaya DSSC sensor has been sensed as V_s , it is amplified as V_{sum} . The V_{in} is as follows:

$$V_{in} = \frac{R_8}{R_7} \times \left(1 + \frac{2R_6}{R_5}\right) \times (V_{sum} - V_{Bias})$$

and

$$V_{sum} = \frac{R_1 \times V_{Bias} + R_2 \times V_s}{R_1 + R_2} \times \left(1 + \frac{R_4}{R_3}\right) \quad (2)$$

where R_1 to R_5 , R_7 , and R_8 is equal to 100 k Ω , and R_6 is 700 k Ω . The V_{Bias} is 1.5 V. The impedance of the pitaya DSSC sensor is generally small, and it is less than 1 k Ω . Thus, the R_1 and R_2 must be larger than 1 k Ω for impedance matching. However, values of resistors still have limitations when they are fabricated into a single chip. Thus, values of 100 k Ω are thus chosen. After the analog front-end circuits have delivered a voltage, the voltage-to-current converter with accuracy must convert that voltage into a current. Next, through a current mirror, the capacitor is to be charged and discharged by a current. The current I_1 is derived as

$$I_1 = k_{p1} (V_{sg1} - V_{t1}) V_{sd1} - \frac{1}{2} V_{sd1}^2$$

and

$$\begin{aligned} I_2 &= \frac{1}{2} k_{p5} (V_{sg5} - V_{t5})^2 \\ &= \frac{1}{2} k_{p5} (V_{s5} - V_{g5} - V_{t5})^2 \\ &= \frac{1}{2} k_{p5} (V_{dd} - V_{g3} - V_{t5})^2 \\ &= \frac{1}{2} k_{p5} [V_{dd} - (V_a - V_{sg3}) - V_{t5}]^2 \\ &= \frac{1}{2} k_{p5} [V_{dd} - (V_a - V_{t3}) - V_{t5}]^2 \\ &= \frac{1}{2} k_{p5} [V_{dd} - (V_{in} - V_{t3}) - V_{t5}]^2 \\ &= \frac{1}{2} k_{p5} (V_{dd} - V_{in} + V_{t3} - V_{t5})^2 \end{aligned}$$

and

$$\begin{aligned} I_{sum} &= I_1 + M_1 I_2 \\ &= [k_{p1} (V_{sg1} - V_{t1}) (V_{dd} - V_{in}) - \frac{1}{2} k_{p1} (V_{dd} - V_{in})^2 \\ &\quad + \frac{1}{2} M_1 k_{p5} (V_{dd} - V_{in} + V_{t3} - V_{t5})^2] \\ &= [k_{p1} (V_{sg1} - V_{t1} - V_{t3} + V_{t5}) V_{dd} - k_{p1} (V_{sg1} - V_{t1} \\ &\quad + V_{t3} - V_{t5}) V_{in} + \frac{1}{2} k_{p1} (V_{t3} - V_{t5})^2] \quad (3) \end{aligned}$$

where M_1 is 1, the V_{t1} , V_{t3} , and V_{t5} are the threshold voltages of MOSFETs M_{1p} , M_{3p} , and M_{5p} . The terms k_{p1} and k_{p5} are the transconductance values of M_{1p} and M_{5p} . If the V_{t3} is equal to V_{t5} , the I_{sum} could be ideally derived as

$$I_{sum} = k_{p1} (V_{sg1} - V_{t1}) (V_{dd} - V_{in}). \quad (4)$$

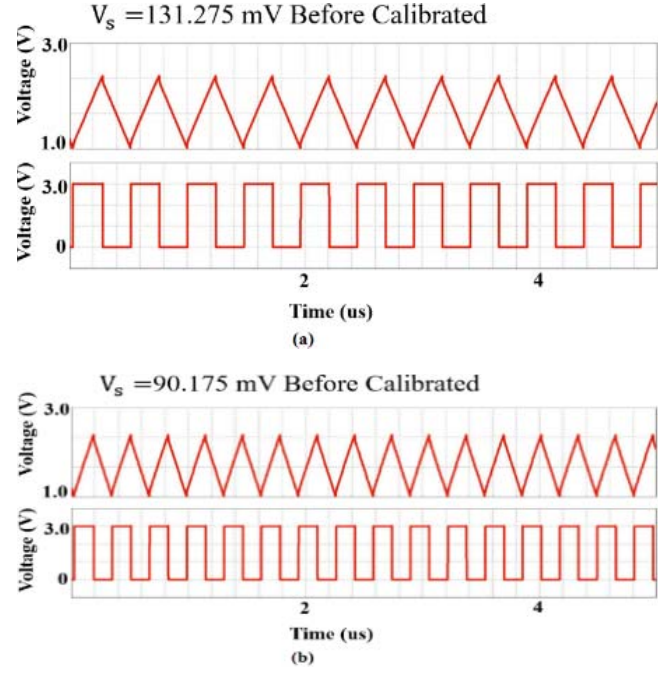


Fig. 3. Simulations performed before calibrations, the V_s of (a) 131.275 and (b) 90.175 mV.

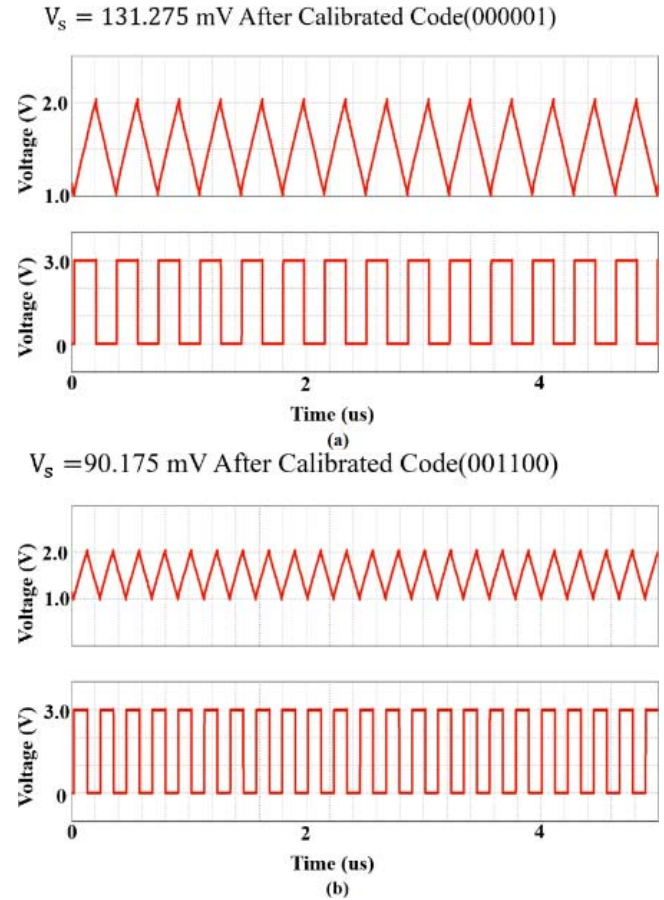


Fig. 4. Simulations performed after calibrations, the V_s of (a) 131.275 and (b) 90.175 mV.

However, the threshold of each MOS is always not the same after integrated circuit fabrication. Thus, V_{t3} and V_{t5} cannot completely cancel each other. The situation requires

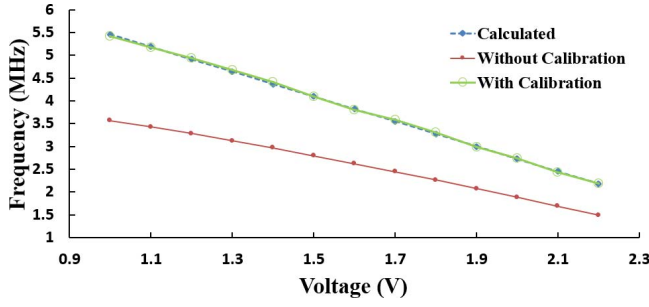


Fig. 5. Simulations of the whole range of V_{in} .

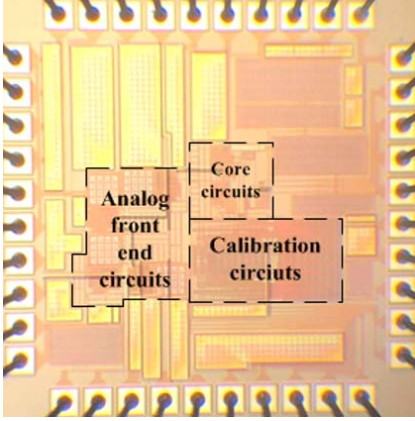


Fig. 6. The photograph of the fabricated chip. The chip's area was $1.49 \times 1.49 \text{ mm}^2$.

a calibration method to wholly eliminate the difference of the V_{I3} and V_{I5} ; this calibration method is discussed later. The current I_{sum} is the input current of the current-to-frequency converter. First, the voltage V_{in} is obtained from the pitaya DSSC sensor. The capacitor C_1 is then initially charged by the I_C . When the voltage on the capacitor C_1 touches the threshold voltage V_H , the RS flip-flop output signals Q and $\bar{Q}$ are logic high and low. The capacitor then begins to be discharged. The voltage on the capacitor C_1 touches the threshold voltage V_L , the RS flip-flop output signals Q and $\bar{Q}$ are converted into logic low and high.

The capacitor is then charged. However, the nonlinear terms of the I_{sum} are still vital. Thus, a calibration's operation is considered. Adding the calibration's operation, the output current of the voltage-to-current converter is derived as

$$I_{sum} = [k_{p1}(V_{sg1} - V_{t1} - V_{t3} + V_{t5})V_{dd} - k_{p1}(V_{sg1} - V_{t1} + V_{t3} - V_{t5})V_{in} + \frac{1}{2}k_{p1}(V_{t3} - V_{t5})^2] + M_3 I_{Cali};$$

$$M_3 = \frac{\left(\frac{W}{L}\right)_{32n}}{\left(\frac{W}{L}\right)_{31n}}$$

$$F_{out} = \frac{[k_{p1}(V_{sg1} - V_{t1} - V_{t3} + V_{t5})V_{dd} - k_{p1}(V_{sg1} - V_{t1} + V_{t3} - V_{t5})V_{in}]}{2C_1(V_H - V_L)} \times \frac{+\frac{1}{2}k_{p1}(V_{t3} - V_{t5})^2 + M_3 I_{Cali}}{2C_1(V_H - V_L)}. \quad (6)$$

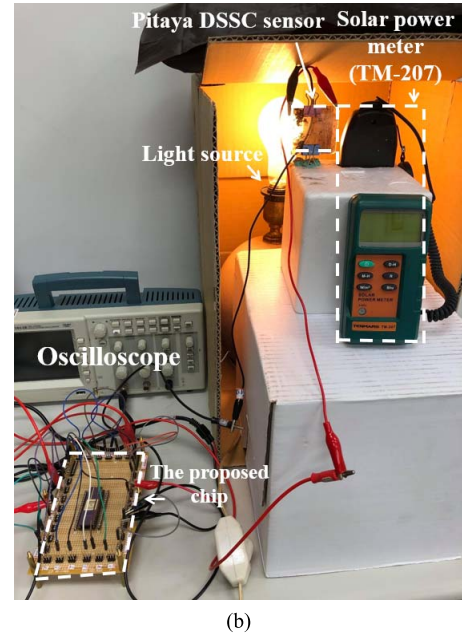
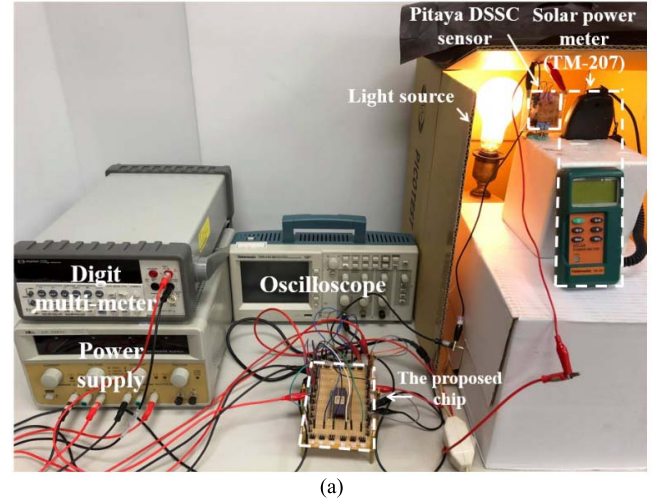


Fig. 7. (a) The demonstration of the whole system. (b) The close view.

and

$$I_{Cali} = I_{CaliA} \times E_n \left(1 + A_0 \times 2^0 + A_1 \times 2^1 + A_2 \times 2^2 + A_3 \times 2^3 + A_4 \times 2^4 + A_5 \times 2^5 \right). \quad (5)$$

The term I_{CaliA} is the unit current. The enable signal E_n is able to start up calibrations. The codes A_0 to A_5 is to control the calibration current I_{Cali} . Finally, the output frequency F_{out} is wholly derived in (6), as shown at the bottom of this page. By the proposed calibration scheme, deviations of output frequency due to threshold differences of each MOSFET can be corrected precisely. Compared with traditional voltage-to-current converters, they do not have a functionality to cancel threshold voltages against each other. Thus, a voltage-to-current conversion with accuracy cannot

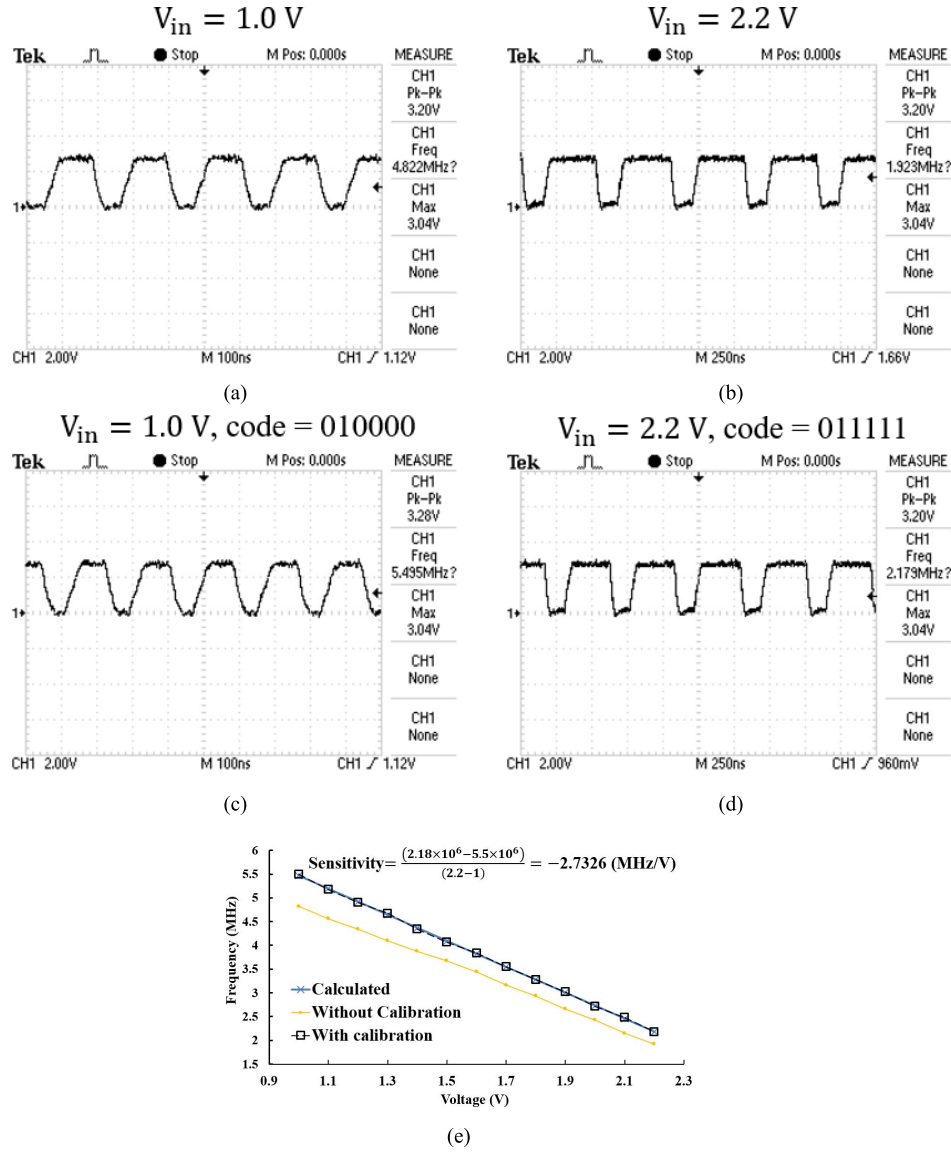


Fig. 8. Measurements of V_{in} at (a) 1.0 V (before) and (c) 1.0 V (after calibration and code = 010000); (b) 2.2 V (before) and (d) 2.2 V (after calibration and code = 011111); (e) 1.0 to 2.2 V.

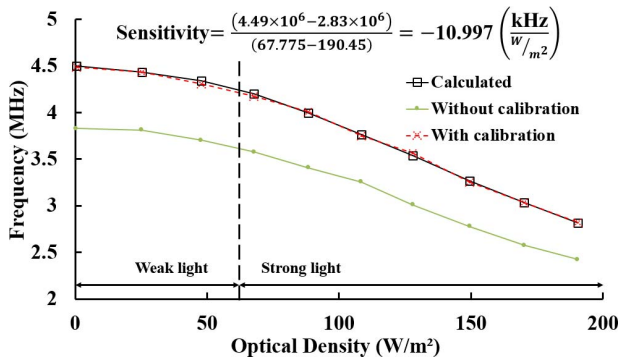


Fig. 9. Measurements of a combination of the pitaya DSSC sensor with the proposed sensor transducer.

be guaranteed. After the entire circuit had been constructed, simulations were conducted. Without and with calibrations, circuit operations are simulated in Fig. 3 and 4. Simulated circuit operations were successfully conducted. As illustrated

	[29]	[30]	This work
Technology	0.25 μ m CMOS	0.35 μ m CMOS	0.35 μ m CMOS
Supply voltage	2.5 V	3.0 V	3.0 V
Sensitivity	520 kHz/V	833.33 kHz/V	-2.73 MHz/V
Input range	0-1.2 V	0-4.8 V	1-2.2 V
Output frequency	52 - 416 kHz	0.62 - 4.03 MHz	2.18 - 5.5 MHz
Error	N/A	< 0.39%	< 0.86%

in Fig. 5, the output frequency linearly corresponded to its input voltage V_{in} . In the later section, measurement results are presented.

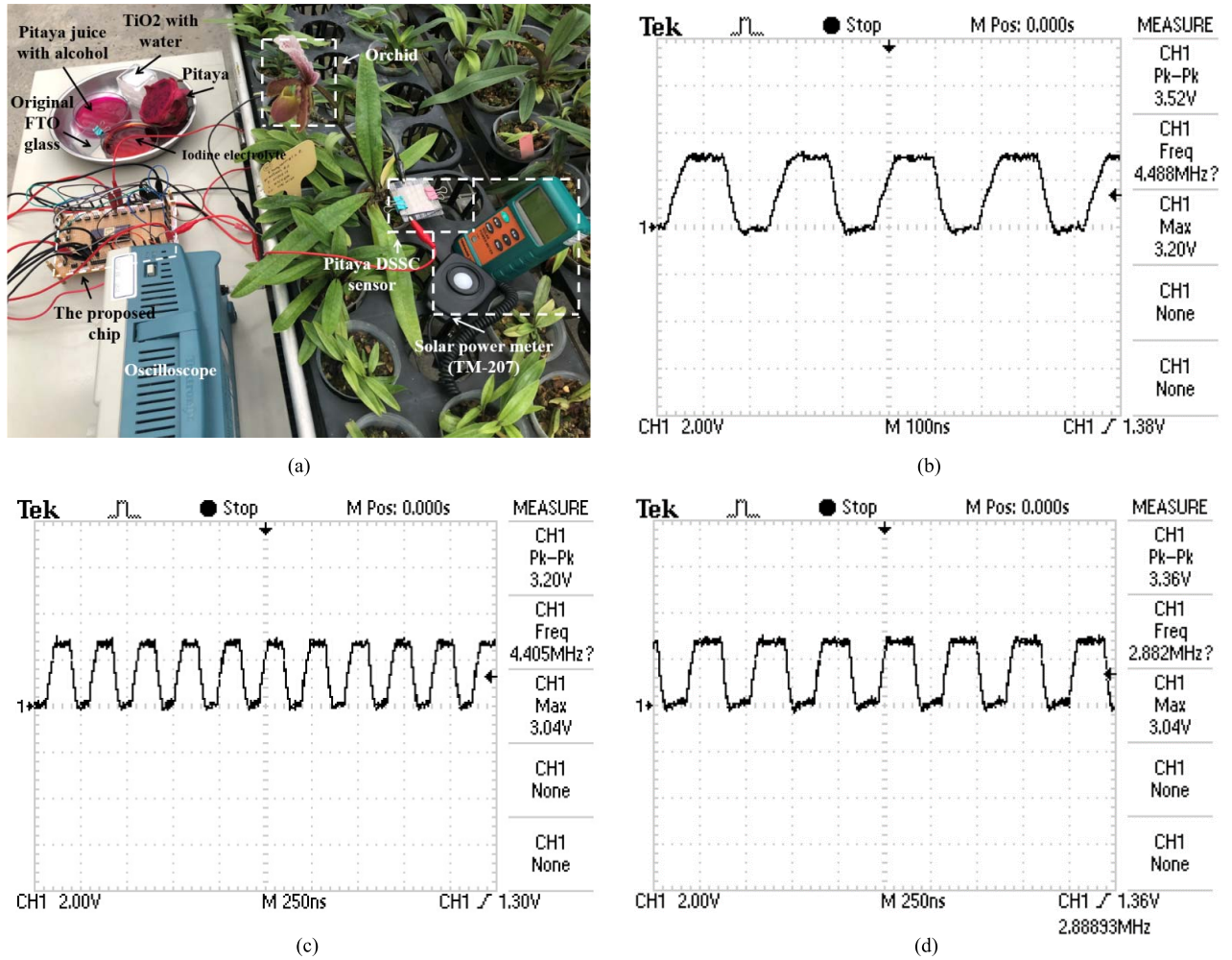


Fig. 10. An experiment for orchid environmental sunlight intensity detection (a) the demonstration, (b) sunlight intensity of 5.4 W/m², (c) sunlight intensity of 33.5 W/m², and (d) sunlight intensity of 160.9 W/m².

III. MEASUREMENT RESULTS

Figure 6 showed the photograph of the fabricated chip. The pads on the left and right sides were connected to analog and digital outputs of the chip. The pads on the top and bottom sides were connected to bias voltages and digital codes of the chip. The chip's area was 1.49×1.49 mm². The whole system, as illustrated in Fig. 7, consisted of power supplies, a pitaya DSSC sensor, a Tektronix TDS-1012B oscilloscope, a PICOTEST digital multimeter M3500A, a solar power meter TM-207, and the proposed chip. The analog output of the pitaya DSSC sensor was gained by the digital multimeter; the solar power meter was used to monitor the sunlight intensity; and the oscilloscope was used to capture the proposed chip's output frequency. First, the proposed chip was verified as depicted in Fig. 8(a)–(e). The input voltage V_{in} ranged from 1.0 to 2.2 V; the corresponding frequency ranged from 5.495 to 2.179 MHz. The output frequency was linearly proportional to its input voltage. The sensitivity was -2.7326 MHz/V. Till now, it was the highest one and the comparison with prior arts were displayed in Table I. A maximum linear error of 0.86% was calculated. After the proposed chip had been accurately calibrated and the nonlinear

terms (caused by the threshold variations of MOSFETs) had been completely eliminated, the proposed chip demonstrated a high sensitivity. A compact combination of the pitaya DSSC sensor and the proposed chip were verified, with the results illustrated in Fig. 9. A sensitivity of -10.997 kHz/(W/m²) was obtained, and a maximum linear error of 0.79% was calculated. Operations of the pitaya DSSC monitor were appropriately validated.

Finally, by using the proposed pitaya DSSC monitor, an experiment was conducted for monitoring orchid environmental sunlight intensity, with the results presented in Fig. 10. The output frequencies of 4.488, 4.405, and 2.882 MHz were corresponding to sunlight intensities of 5.4, 33.5, and 160.9 W/m², as graphed in Fig. 8. By using the TM-207 solar power meter, sunlight intensities of 7.7, 36.8, and 163.1 W/m² were also detected respectively. Both light intensity values were close, and thus represented a compact combination of the pitaya DSSC sensor and the sensor transducer was successfully functioned. Backend digital systems, such as IoT devices, could thus utilize the output frequency to inform flower farmers about the intensity of sunlight illuminating their orchids whenever necessary. Moreover, other DSSC sensors could also easily be integrated with the proposed chip due to

TABLE II
SUMMARIZATION OF THE PROPOSED PITAYA DSSC MONITOR

Technology	TSMC 0.35 μm CMOS 2P4M
Power supply	3.0 V
With calibrations	Provided
Sensitivity of the proposed chip	-2.7326 (MHz/V)
Maximum linear error (after calibration)	0.86 %
Input range of optical density of pitaya DSSC sensor	67.775 to 190.45 (W/m^2)
Corresponding output frequency range	2.83 to 4.49 MHz
Sensitivity of the whole system	-10.997 ($\frac{\text{kHz}}{\text{W}/\text{m}^2}$)
Maximum linear error (after calibration)	0.79 %
Chip physical layout area	1.49 x 1.49 mm^2
Application field	Sunlight intensity detection in greenhouse

functionality of calibrations. A summarization of the pitaya DSSC monitor is displayed in Table II.

IV. CONCLUSION

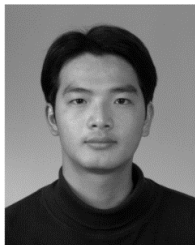
A pitaya DSSC monitor for sunlight intensity detecting was proposed. A compact combination of the pitaya DSSC sensor and sensor transducer was well functioned. The results reported that circuits could be utilized in detecting different levels of sunlight intensity. A validation of functionalities of the proposed pitaya DSSC monitor were successfully measured. In the future, for monitoring sunlight intensity in greenhouses, the proposed chip will be combined with IoT devices.

ACKNOWLEDGMENT

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